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Copper is an essential element, and toxic effects are not common. However, given sufficient exposure to high levels by inhalation, absorption, or ingestion from sources such as industrial hazards or contaminated groundwater, copper toxicity can occur.

Salts that are more soluble tend to exhibit higher toxicity when ingested, but the solubilities of copper salts vary and are also temperature-dependent, as shown by selected examples in Figure 1.

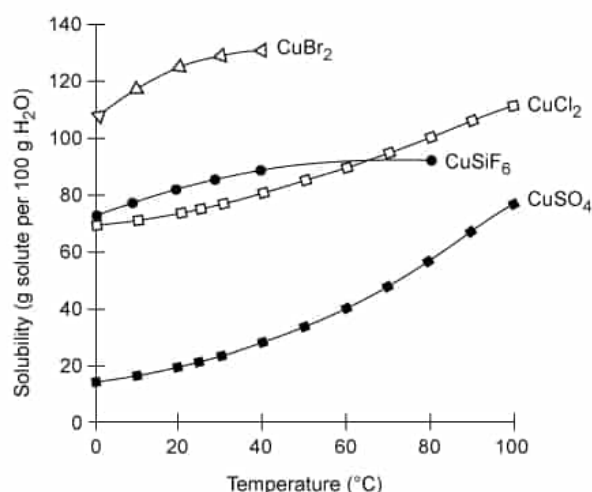
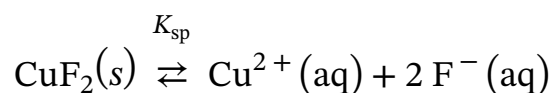


Figure 1 The solubility of selected copper salts in distilled water with increasing temperature

Some copper salts have solubilities much lower than those shown in Figure 1. For example, CuF₂ has a solubility of only 0.075 g per 100 g of water (at 25 °C). This indicates that CuF₂ dissociates in water only to a small extent (Reaction 1) and that the equilibrium favors the undissociated salt, giving CuF₂ a solubility product constant of $K_{sp} = 1.6 \times 10^{-6}$.

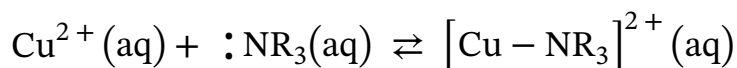


Reaction 1

Compound-specific properties can also impact solubility. In CuF₂, the dissociated F⁻ ion is mildly basic ($pK_b = 10.8$), causing the equilibrium to be influenced by the pH of the solution.

Specific characteristics of copper can also play a role. For example, copper ions in

solution express an affinity to form coordinate bonds with available lone-pair electrons of nitrogen atoms in amines (Reaction 2).



Reaction 2

If excess amine ligands are present, coordinate bonds like that shown in Reaction 2 will continue to form until each Cu^{2+} ion in solution is coordinated with four nitrogen atoms. This acquisition of copper ions by ligands in solution can also impact the solubility equilibrium of copper salts by forming soluble coordination complexes.

A saturated solution of CuF_2 dissolved in pure water at 25°C has a Cu^{2+} ion concentration of $7.4 \times 10^{-3} \text{ M}$. If measured at the same temperature, what is the Cu^{2+} ion concentration in a saturated solution of CuF_2 dissolved in water that also contains 0.20 M NaF ?

- ☐ A. $8.1 \times 10^{-6} \text{ M}$ [13%]
- ✓ ☒ B. $4.0 \times 10^{-5} \text{ M}$ [51%]
- ☐ C. $7.4 \times 10^{-3} \text{ M}$ [21%]
- ☐ D. $6.4 \times 10^{-2} \text{ M}$ [13%]

Correct

50% answered correctly

Explanation:

Calculating the solubility of CuF_2 in 0.20 M NaF

	$\text{CuF}_2(\text{s}) \xrightleftharpoons{K_{\text{sp}}} \text{Cu}^{2+}(\text{aq}) + 2\text{F}^{-}(\text{aq})$		
Initial	excess	0	$(0 + 0.20)$
Change	$-x$	$+x$	$+2x$
Equilibrium	excess $- x$	$+x$	$(0.20 + 2x)$

0 from CuF_2
0.20 from NaF

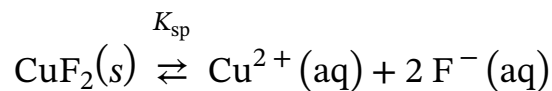
If x is small,
 $(0.20 + 2x) \approx 0.20$

$$K_{\text{sp}} = [\text{Cu}^{2+}][\text{F}^{-}]^2 = (x)(0.20 + 2x)^2 \approx (x)(0.20)^2$$

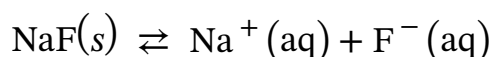
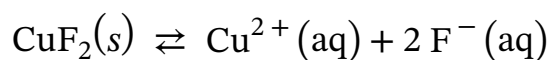
$$x = 4.0 \times 10^{-5} \text{ M}$$

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Solubility refers to the amount of a compound that can dissolve per unit volume of a solvent at a particular temperature. A **saturated solution** contains the maximum amount of solute that can dissolve under the given conditions. When an ionic compound is placed in water, the composite ions will **dissociate** to some extent into the solution and establish an **equilibrium**. The solubility product constant K_{sp} of the equilibrium provides an indication of the solubility of the dissolved compound.



If a solution contains two salts that have an **ion in common**, the common ion supplied by one salt produces a **common ion effect** that stresses the product side of the equilibrium of the other salt and **shifts the equilibrium** to the **left** (ie, toward the undissolved state). Accordingly, for both salts the **solubilities decrease**.



When dissolving CuF_2 into water that contains 0.20 M NaF, initially there is no Cu^{2+} or F^{-} in solution from CuF_2 , but there is 0.20 M F^{-} from the NaF. If an amount x of CuF_2 subsequently dissolves in the solution, the changes and equilibrium concentrations can be evaluated as shown in the table above. Placing these equilibrium values into the K_{sp} expression for CuF_2 gives:

$$K_{\text{sp}} = [\text{Cu}^{2+}][\text{F}^{-}]^2 = (x)(0.20 + 2x)^2$$

Because the K_{sp} of CuF_2 is small, it can be assumed that x is very small compared to 0.20 M. As such, $(0.20 + 2x)$ can be approximated as 0.20 M to simplify the calculation:

$$K_{\text{sp}} = (x)(0.20 + 2x)^2 \approx (x)(0.20)^2 = 0.040x$$

As stated in the passage, the K_{sp} of CuF_2 is 1.6×10^{-6} . Substituting this value of the K_{sp} into the equation and solving for x gives:

$$1.6 \times 10^{-6} = 0.040x$$

$$x = 4.0 \times 10^{-5} \text{ M Cu}^{2+}$$

Therefore, in a saturated solution of CuF_2 that also contains 0.20 M NaF, the concentration of Cu^{2+} ions is $4.0 \times 10^{-5} \text{ M}$, which is less than its concentration in pure water.

(Choice A) This answer results from not squaring the concentration term (0.20 M).

(Choice C) The common ion effect will cause the concentration of Cu^{2+} ions to be lower in 0.20 M NaF than in pure water. The solubility cannot be the same in both solutions.

(Choice D) Both solutions are aqueous, and there are no additional factors present

to influence the solubility equilibrium. Therefore, the concentration cannot exceed that observed for pure water.

Educational objective:

In solutions containing two salts that have an ion in common, the common ion supplied by the one salt produces a common ion effect that shifts the equilibrium of the other salt to the left and decreases its solubility.

Time spent:0 sec

Subject:
General Chemistry

QID:402238

System:
Solutions and Electrochemistry

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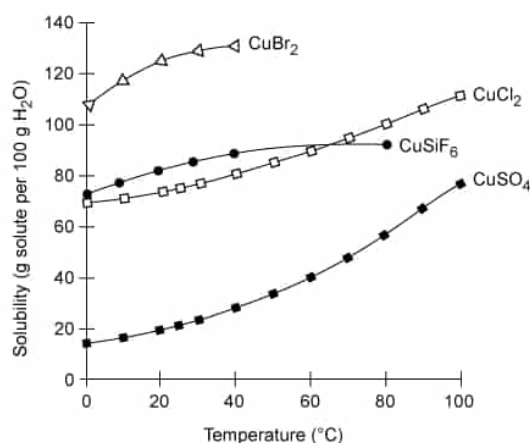
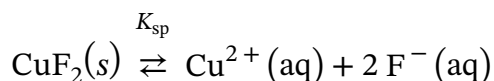


Figure 1 The solubility of selected copper salts in distilled water with increasing temperature

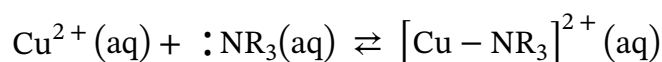
Some copper salts have solubilities much lower than those shown in Figure 1. For example, CuF₂ has a solubility of only 0.075 g per 100 g of water (at 25 °C). This indicates that CuF₂ dissociates in water only to a small extent (Reaction 1) and that the equilibrium favors the undissociated salt, giving CuF₂ a solubility product constant of $K_{sp} = 1.6 \times 10^{-6}$.



Reaction 1

Compound-specific properties can also impact solubility. In CuF₂, the dissociated F[−] ion is mildly basic ($pK_b = 10.8$), causing the equilibrium to be influenced by the pH of the solution.

Specific characteristics of copper can also play a role. For example, copper ions in solution express an affinity to form coordinate bonds with available lone-pair electrons of nitrogen atoms in amines (Reaction 2).



Reaction 2

If excess amine ligands are present, coordinate bonds like that shown in Reaction 2 will continue to form until each Cu²⁺ ion in solution is coordinated with four nitrogen atoms. This

acquisition of copper ions by ligands in solution can also impact the solubility equilibrium of copper salts by forming soluble coordination complexes.

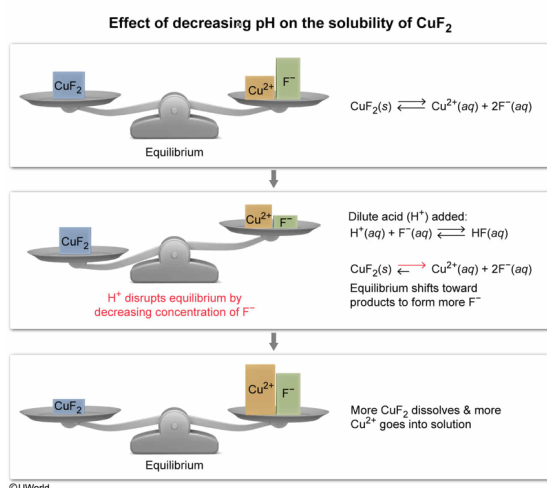
Which of the following changes to the solution will increase the amount of $\text{CuF}_2(\text{s})$ that dissolves and raise the $\text{Cu}^{2+}(\text{aq})$ concentration in the solution?

- ✔ ☒ A. Adding dilute nitric acid (a source of H^+ ions) [59%]
☐ B. Adding dilute aqueous sodium hydroxide ($\text{Cu}(\text{OH})_2$ has a $K_{\text{sp}} = 2.2 \times 10^{-20}$) [22%]
☐ C. Diluting the solution with aqueous potassium fluoride [9%]
☐ D. Evaporating some solution to decrease the solution volume [9%]

Correct

59% answered correctly

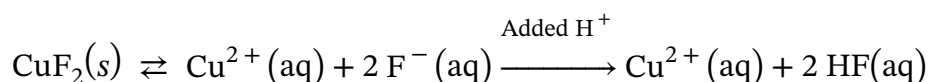
Explanation:



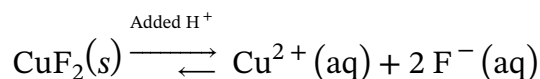
According to **Le Châtelier's principle**, if an equilibrium is disrupted, it will shift to minimize the change and establish a new equilibrium. For a chemical reaction at equilibrium, this principle indicates that if some of the products of the reaction are removed, the equilibrium will shift to form more of the products.

As discussed in the passage, the solubility of $\text{CuF}_2(\text{s})$ in water is low because the equilibrium (Reaction 1) favors the undissolved salt. However, the fluoride ion is mildly **basic**, and this property can be exploited to **disrupt the equilibrium** because bases react with **acids**.

Therefore, if nitric acid (a source of H^+) is added to the solution, some of the $\text{F}^{-}(\text{aq})$ from the dissolved $\text{CuF}_2(\text{s})$ will be used by the acid to form HF.



Because $\text{HF}(\text{aq})$ is a weak acid, it only partially ionizes in water, and more of the $\text{F}^{-}(\text{aq})$ gets taken into solution and bound as $\text{HF}(\text{aq})$. This causes the concentration of $\text{F}^{-}(\text{aq})$ from the dissolved $\text{CuF}_2(\text{s})$ to initially decrease. The temporary decrease in $\text{F}^{-}(\text{aq})$ disrupts the $\text{CuF}_2(\text{s})$ equilibrium, causing a **shift toward the products** to form more $\text{F}^{-}(\text{aq})$.



Consequently, more $\text{CuF}_2(s)$ will dissolve and the $\text{Cu}^{2+}(aq)$ concentration will increase on adding dilute nitric acid.

(Choice B) Although OH^- ions (from sodium hydroxide) disturbs the equilibrium by reacting with some of the Cu^{2+} ions to form $\text{Cu}(\text{OH})_2$, the very small K_{sp} of $\text{Cu}(\text{OH})_2$ indicates that it is very insoluble in water. Therefore, adding OH^- ions would convert some $\text{CuF}_2(s)$ to $\text{Cu}(\text{OH})_2(s)$ (ie, convert one solid to another) and precipitate Cu^{2+} ions from the solution as $\text{Cu}(\text{OH})_2(s)$. These reactions would *lower* the $\text{Cu}^{2+}(aq)$ concentration.

(Choice C) Diluting the solution with $\text{KF}(aq)$ will increase the $\text{F}^-(aq)$ concentration. Due to the common ion effect, this will shift the equilibrium of Reaction 1 toward the reactants, thereby decreasing the amount of $\text{CuF}_2(s)$ that dissolves and lowering the $\text{Cu}^{2+}(aq)$ concentration.

(Choice D) The maximum amount of $\text{CuF}_2(s)$ that will dissolve in pure water remains constant at 0.075 g CuF_2 per 100 g H_2O . As the water evaporates, some dissolved CuF_2 will precipitate, but the $\text{Cu}^{2+}(aq)$ concentration per unit volume will remain unchanged.

Educational objective:

Changing the pH of a solution can enhance the solubility of salts that have an acidic or basic ion. This occurs by applying Le Châtelier's principle to disrupt the solubility equilibrium and shift the reaction toward the products.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402239

System:
Solutions and Electrochemistry

Copper plays a vital role as a trace mineral for biological processes. The amount of dietary copper needed by an adult each day is about 30 µg/kg of body weight, and blood plasma normally has a copper concentration of 1 mg/L.

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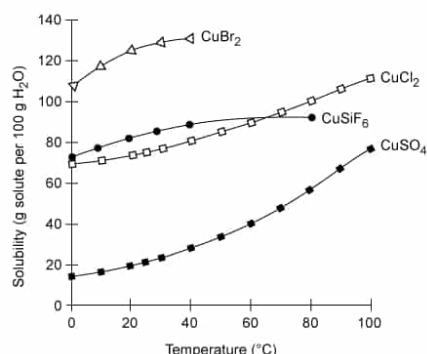
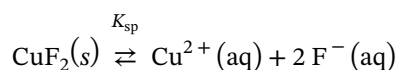


Figure 1 The solubility of selected copper salts in distilled water with increasing temperature

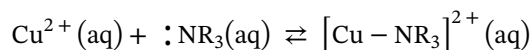
Some copper salts have solubilities much lower than those shown in Figure 1. For example, CuF₂ has a solubility of only 0.075 g per 100 g of water (at 25 °C). This indicates that CuF₂ dissociates in water only to a small extent (Reaction 1) and that the equilibrium favors the undissociated salt, giving CuF₂ a solubility product constant of $K_{sp} = 1.6 \times 10^{-6}$.



Reaction 1

Compound-specific properties can also impact solubility. In CuF₂, the dissociated F[−] ion is mildly basic ($pK_b = 10.8$), causing the equilibrium to be influenced by the pH of the solution.

Specific characteristics of copper can also play a role. For example, copper ions in solution express an affinity to form coordinate bonds with available lone-pair electrons of nitrogen atoms in amines (Reaction 2).



Reaction 2

If excess amine ligands are present, coordinate bonds like that shown in Reaction 2 will continue to form until each Cu²⁺ ion in solution is coordinated with four nitrogen atoms. This acquisition of copper ions by ligands in solution can also impact the solubility equilibrium of copper salts by forming soluble coordination complexes.

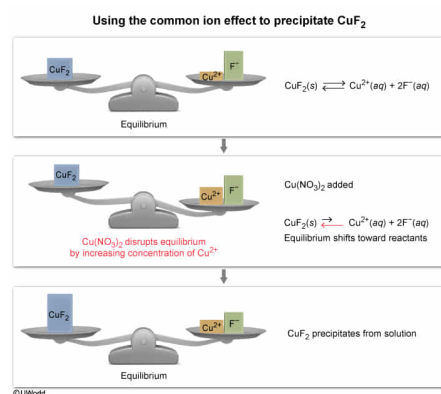
Suppose that a research technician wants to separate an aqueous mixture of CuF₂ and BaF₂ ($K_{sp} = 3.0 \times 10^{-6}$) by precipitating CuF₂ from the solution. What should be added to the solution to perform the separation?

- ✓ ☒ A. Cu(NO₃)₂ [51%]
☐ B. NaF [19%]
☐ C. Ba(NO₃)₂ [21%]
☐ D. H₂O [7%]

Correct

52% answered correctly

Explanation:



If a solution contains two salts that have an **ion in common**, the common ion supplied by one salt introduces a **common ion effect** that disturbs the solubility equilibrium of the other salt and causes **its equilibrium** to shift to the left (toward the reactants), thereby decreasing the solubility of the other salt. The common ion effect is essentially a specific scenario involving **Le Châtelier's principle**, which states that if an equilibrium is disrupted, the equilibrium will shift to minimize the change and establish a new equilibrium.

In some instances, these concepts can be applied in the lab to achieve the separation of a mixture by selective precipitation of a desired compound in a solution. To separate a solution mixture of CuF_2 and BaF_2 by precipitating CuF_2 , Cu^{2+} ions can be introduced into the solution. The increased Cu^{2+} concentration disturbs the product side of the solubility equilibrium and causes it to **shift toward the reactants** (undissolved salt). Therefore, adding $\text{Cu}(\text{NO}_3)_2$ as a source of Cu^{2+} ions will selectively cause the CuF_2 in the mixture to precipitate.

(Choice B) Adding NaF would increase the fluoride ion concentration in the mixture. Because CuF_2 and BaF_2 both have fluoride anions, this would decrease the solubility of both compounds and cause some of both to precipitate together, making the separation unsuccessful.

(Choice C) The addition of $\text{Ba}(\text{NO}_3)_2$ would increase the Ba^{2+} ion concentration in the mixture. The common ion effect would cause BaF_2 to precipitate rather than CuF_2 .

(Choice D) Adding more water to the mixture would not assist precipitation of either compound. This would have the opposite effect because more of the compounds can dissolve with an increased volume of water.

Educational objective:

In solutions containing two salts that have an ion in common, the common ion supplied by one salt produces a common ion effect that shifts the equilibrium of the other salt to the left and decreases its solubility. The common ion effect is a specific scenario involving Le Châtelier's principle that can be used to separate compound mixtures.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402240

System:
Solutions and Electrochemistry

Copper plays a vital role as a trace mineral for biological processes. The amount of dietary copper needed by an adult each day is about 30 µg/kg of body weight, and blood plasma normally has a copper concentration of 1 mg/L.

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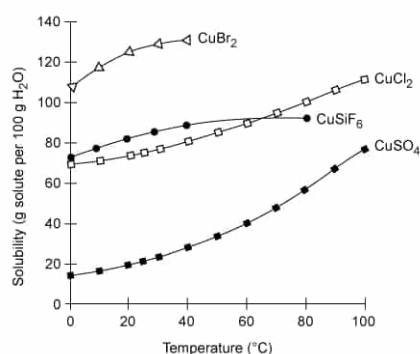
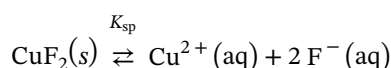


Figure 1 The solubility of selected copper salts in distilled water with increasing temperature

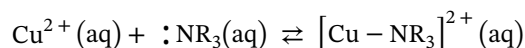
Some copper salts have solubilities much lower than those shown in Figure 1. For example, CuF₂ has a solubility of only 0.075 g per 100 g of water (at 25 °C). This indicates that CuF₂ dissociates in water only to a small extent (Reaction 1) and that the equilibrium favors the undissociated salt, giving CuF₂ a solubility product constant of $K_{sp} = 1.6 \times 10^{-6}$.



Reaction 1

Compound-specific properties can also impact solubility. In CuF₂, the dissociated F⁻ ion is mildly basic ($pK_b = 10.8$), causing the equilibrium to be influenced by the pH of the solution.

Specific characteristics of copper can also play a role. For example, copper ions in solution express an affinity to form coordinate bonds with available lone-pair electrons of nitrogen atoms in amines (Reaction 2).



Reaction 2

If excess amine ligands are present, coordinate bonds like that shown in Reaction 2 will continue to form until each Cu²⁺ ion in solution is coordinated with four nitrogen atoms. This acquisition of copper ions by ligands in solution can also impact the solubility equilibrium of copper salts by forming soluble coordination complexes.

During the separation of a fully dissolved mixture of CuF₂ ($K_{sp} = 1.6 \times 10^{-6}$) and BaF₂ ($K_{sp} = 3.0 \times 10^{-6}$) from a saturated aqueous solution, a lab technician chose to use the common ion effect to precipitate CuF₂ from the solution rather than BaF₂. Which of the following statements does NOT explain why precipitating CuF₂ is the better choice for the separation procedure?

- ☐ A. CuF₂ is less soluble and more responsive to small changes in common ion concentrations. [11%]
- ☒ B. Adding F⁻ as the common ion would cause CuF₂ to precipitate. [48%]
- ☐ C. There are fewer moles of Cu²⁺(aq) to remove from the solution than moles of Ba²⁺(aq). [29%]
- ☐ D. Precipitating BaF₂ would require a greater number of common ions to be added to the solution. [10%]

Explanation:

Choosing a compound to precipitate by K_{sp}

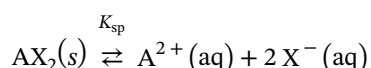


$\text{CuF}_2(s) \rightleftharpoons \text{Cu}^{2+}(aq) + 2\text{F}^{-}(aq)$
 $K_{sp} = [\text{Cu}^{2+}][\text{F}^{-}]^2 = 1.6 \times 10^{-6}$

- Less soluble (smaller K_{sp})
- Fewer moles to remove
- More responsive to changing $[\text{Cu}^{2+}]$ or $[\text{F}^{-}]$
- Fewer common ions must be added to induce precipitation

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When an ionic compound AX_2 is placed in water, the composite ions A^{2+} and X^{-} will dissociate to some extent into the solution and establish an equilibrium in which the rate of dissolution (forward reaction) equals the rate of precipitation (reverse reaction).



This equilibrium position is indicated by the **solubility product constant** K_{sp} , which provides a measure of the **solubility** of a compound at a given temperature.

$$K_{sp} = [\text{A}^{2+}][\text{X}^{-}]^2$$

Compounds with larger K_{sp} values are more soluble than compounds with smaller K_{sp} values.

Because the K_{sp} of CuF_2 (1.6×10^{-6}) is smaller than the K_{sp} of BaF_2 (3.0×10^{-6}), a saturated aqueous mixture of CuF_2 and BaF_2 will have fewer moles of $\text{Cu}^{2+}(aq)$ present in the solution to remove during a separation (**Choice C**). This can be demonstrated mathematically by the ratio of the K_{sp} values as:

$$\begin{aligned} \frac{K_{sp}(\text{BaF}_2)}{K_{sp}(\text{CuF}_2)} &= \frac{[\text{Ba}^{2+}][\text{F}^{-}]^2}{[\text{Cu}^{2+}][\text{F}^{-}]^2} \\ \frac{3.0 \times 10^{-6}}{1.6 \times 10^{-6}} &= \frac{[\text{Ba}^{2+}][\text{F}^{-}]^2}{[\text{Cu}^{2+}][\text{F}^{-}]^2} \\ 1.9 &= \frac{[\text{Ba}^{2+}]}{[\text{Cu}^{2+}]} \end{aligned}$$

This ratio shows that $[\text{Ba}^{2+}]$ is nearly 2 times greater than $[\text{Cu}^{2+}]$ in the solution.

The K_{sp} of a compound is constant at a given temperature, and the concentrations of dissolved species cannot increase beyond that consistent with the K_{sp} . This means that if $[\text{A}^{2+}]$ increases, $[\text{X}^{-}]$ must decrease so that the value of the K_{sp} remains unchanged. If the K_{sp} is small, then small changes to $[\text{A}^{2+}]$ or $[\text{X}^{-}]$ have a much larger impact than would be the case if the K_{sp} were large. Therefore, CuF_2 (smaller K_{sp}) is more responsive than BaF_2 (larger K_{sp}) to small changes in common ion concentrations (**Choice A**), and fully precipitating BaF_2 would require a *greater number* of common ions to be added to the solution (**Choice D**).

As such, Choices A, C, and D are all reasons why precipitating CuF_2 is the better choice for the separation procedure, but the question asks which statement does NOT explain why it is the better choice. Although adding F^{-} as the common ion would cause CuF_2 to precipitate, it would also cause BaF_2 to precipitate. Therefore, the fact that adding F^{-} causes CuF_2 to precipitate does NOT indicate *why* precipitating CuF_2 is a better choice than precipitating BaF_2 (**Choice B**).

Educational objective:

The solubility of a compound is indicated by the solubility product constant K_{sp} , and the K_{sp} values of two compounds can inform which compound to selectively precipitate during the separation of a compound mixture using the common ion effect.

Time spent:0 sec
Subject:
General Chemistry

QID:402241
System:
Solutions and Electrochemistry

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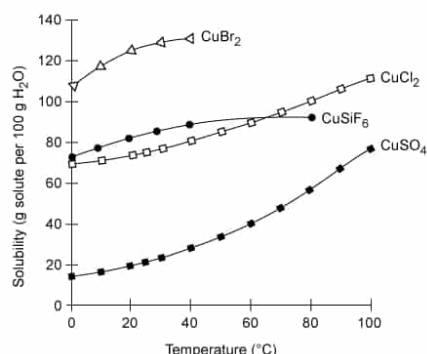
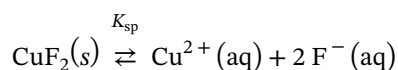


Figure 1 The solubility of selected copper salts in distilled water with increasing temperature

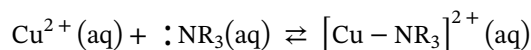
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Reaction 1

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Reaction 2

If excess amine ligands are present, coordinate bonds like that shown in Reaction 2 will continue to form until each Cu²⁺ ion in solution is coordinated with four nitrogen atoms. This acquisition of copper ions by ligands in solution can also impact the solubility equilibrium of copper salts by forming soluble coordination complexes.

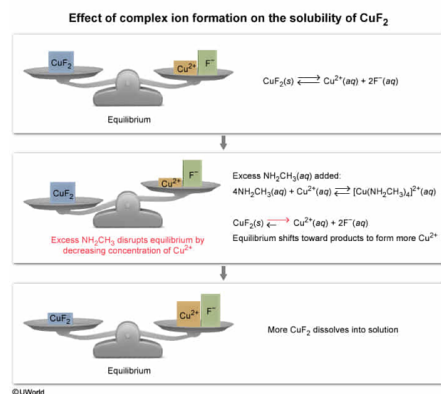
If excess methylamine (CH₃NH₂) is added to a saturated aqueous CuF₂ solution that contains some undissolved CuF₂(s) at the bottom of the solution, the amount of the undissolved CuF₂ that remains will:

- ☐ A. remain the same. [18%]
- ✓ ☒ B. decrease. [62%]
- ☐ C. increase. [11%]
- ☐ D. be unrelated to the amount of CH₃NH₂. [6%]

Correct

63% answered correctly

Explanation:



According to **Le Châtelier's principle**, if an equilibrium is disrupted, it will shift to minimize the change and establish a new equilibrium. For a chemical reaction at equilibrium, this principle indicates that if some of the products of the reaction are removed, the equilibrium will shift to form more of the products.

As discussed in the passage, copper ions in solution express an affinity to form coordinate bonds with available lone-pair electrons of nitrogen atoms in amines (Reaction 2). If excess amine ligands are present with $\text{Cu}^{2+}(\text{aq})$ ions, each Cu^{2+} ion in solution will form coordinate bonds with four nitrogen atoms, as shown above.

The acquisition of $\text{Cu}^{2+}(\text{aq})$ ions by the methylamine ligands forms a **soluble coordination complex** and **disrupts the solubility equilibrium** of CuF_2 by causing the $\text{Cu}^{2+}(\text{aq})$ ion concentration to decrease. The decrease in $\text{Cu}^{2+}(\text{aq})$ will cause the solubility equilibrium to **shift toward the products** to form more $\text{Cu}^{2+}(\text{aq})$. Consequently, on adding excess methylamine, the amount of undissolved $\text{CuF}_2(\text{s})$ that remains will decrease as more dissolves into the solution.

(Choices A, C, and D) In the presence of excess amines, $\text{Cu}^{2+}(\text{aq})$ ions form coordination complexes, each with four coordinate bonds to nitrogen atoms. Because the amines use available $\text{Cu}^{2+}(\text{aq})$ ions to form complexes, the solubility equilibrium of $\text{CuF}_2(\text{s})$ will be disturbed and will shift to the right (toward products) to replace the used $\text{Cu}^{2+}(\text{aq})$ ions.

As a result, the amount of $\text{CuF}_2(\text{s})$ that dissolves will *increase*.

Educational objective:

Adding ligands that can form soluble coordination complexes with metal ions in a solution can enhance the solubility of ionic salts. This occurs by applying Le Châtelier's principle to disrupt the solubility equilibrium and shift the reaction toward the products.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402242

System:
Solutions and Electrochemistry

Copper plays a vital role as a trace mineral for biological processes. The amount of dietary copper needed by an adult each day is about 30 µg/kg of body weight, and blood plasma normally has a copper concentration of 1 mg/L.

Copper is an essential element, and toxic effects are not common. However, given sufficient exposure to high levels by inhalation, absorption, or ingestion from sources such as industrial hazards or contaminated groundwater, copper toxicity can occur.

Salts that are more soluble tend to exhibit higher toxicity when ingested, but the solubilities of copper salts vary and are also temperature-dependent, as shown by selected examples in Figure 1.

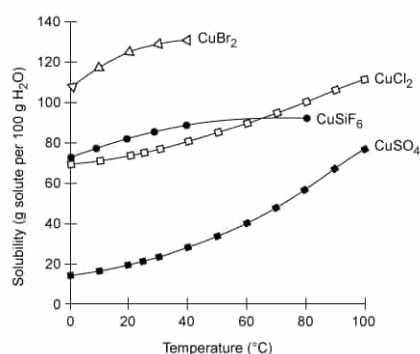
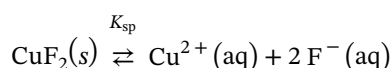


Figure 1 The solubility of selected copper salts in distilled water with increasing temperature

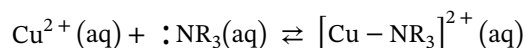
Some copper salts have solubilities much lower than those shown in Figure 1. For example, CuF₂ has a solubility of only 0.075 g per 100 g of water (at 25 °C). This indicates that CuF₂ dissociates in water only to a small extent (Reaction 1) and that the equilibrium favors the undissociated salt, giving CuF₂ a solubility product constant of $K_{sp} = 1.6 \times 10^{-6}$.



Reaction 1

Compound-specific properties can also impact solubility. In CuF₂, the dissociated F⁻ ion is mildly basic ($pK_b = 10.8$), causing the equilibrium to be influenced by the pH of the solution.

Specific characteristics of copper can also play a role. For example, copper ions in solution express an affinity to form coordinate bonds with available lone-pair electrons of nitrogen atoms in amines (Reaction 2).



Reaction 2

If excess amine ligands are present, coordinate bonds like that shown in Reaction 2 will continue to form until each Cu²⁺ ion in solution is coordinated with four nitrogen atoms. This acquisition of copper ions by ligands in solution can also impact the solubility equilibrium of copper salts by forming soluble coordination complexes.

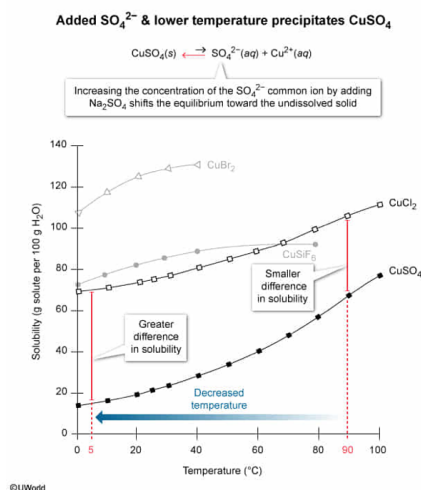
Based on the passage, which of the following procedures would assist in separating an aqueous mixture of CuCl₂ and CuSO₄ salts that is initially at 90 °C?

- I. Cool the solution mixture to 5°C
 - II. Add Na₂SO₄ to the solution mixture
 - III. Lower the pH of the mixture
- ☐ A. I only [9%]
☐ B. II only [16%]
☒ C. I and II only [50%]
☐ D. I, II, and III only [23%]

Correct

49% answered correctly

Explanation:



When an ionic compound is placed in water, the composite ions will **dissociate** to some extent into the solution and establish an equilibrium. The **solubility product constant** K_{sp} of the equilibrium provides an indication of the **solubility** of the compound. At a given temperature, the K_{sp} is constant, but changing the temperature alters the solubility equilibrium and its associated K_{sp} .

As seen in Figure 1 in the passage, the solubilities of CuCl_2 and CuSO_4 salts both decrease as the temperature decreases, but the solubility of CuSO_4 decreases to a much greater extent. Consequently, during the separation of a mixture of CuCl_2 and CuSO_4 salts, cooling the mixture from at 90 °C to 5 °C **increases the solubility difference** between the salts, which assists the separation (**Number I**).

To further assist in separating the compounds, the **common ion effect** could be applied by adding Na_2SO_4 to the solution (**Number II**). Because CuSO_4 and Na_2SO_4 have the SO_4^{2-} ion in common, adding Na_2SO_4 will increase the SO_4^{2-} concentration and shift the solubility **equilibrium** toward CuSO_4 to **selectively precipitate** the CuSO_4 out of the mixture.

(**Number III**) Lowering the pH of the mixture by adding an acid would be unhelpful in the separation because Cl^- and SO_4^{2-} are not basic enough to react with the acid and shift the solubility equilibrium of CuCl_2 or CuSO_4 .

Educational objective:

The temperature dependence of solubility and the common ion effect from relevant ions on the solubility equilibrium can be used to assist in selectively precipitating compounds during separations of mixtures.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402243
System:
Solutions and Electrochemistry